Explicit dominant map to the Klein cubic

G = PSL(2,F_11), Phi : P(A^4)=P^19 --> X={sum_j y_j^2 y_{j+1}=0} in P^4.

Coordinates: a=(a_ij), 0<=i<=3, 0<=j<=4. Put B_i=M(a_i), with
M(y)=[[0,-y1,-y3,-y2,-y0,-y4],[y1,0,-y0,0,0,y3],[y3,y0,0,0,-y2,0],
[y2,0,0,0,y4,-y1],[y0,0,y2,-y4,0,0],[y4,-y3,0,y1,0,0]]. Pf(M)=F.
For rst with r<s<t, set L_i[h,rst]=delta_hr(B_i)_st-delta_hs(B_i)_rt+delta_ht(B_i)_rs.
Order J=(012,013,014,015,023,024,025,034,035,045,123,124,125,134,135,145,234,235,245,345).
Stack L_0,L_1,L_2 and write C=(N|r|s), N=first 18 columns, Delta=det N.
For first 18 columns J: u_J=-det N[J<-r], v_J=-det N[J<-s]; set (u245,u345)=(Delta,0), (v245,v345)=(0,Delta).
pi(w)=w012*w034-w013*w024+w014*w023; q0=pi(u), q1=pi(u+v)-pi(u)-pi(v), q2=pi(v).
For eps=0,1 let w^(0)=u,w^(1)=v and D^(eps)[h,i]=sum_{k,J}(B_i)[h,k](L_3)[k,J]w^(eps)_J, h=0,1,2.
Delete column i: E_i from D^(0), H_i from D^(1). Define
(c_i0,c_i1,c_i2,c_i3)=(-1)^i(det E_i, tr(adj(E_i)H_i), tr(adj(H_i)E_i), det H_i).
For j=0,...,4:
P_j=sum_i a_ij(q2^2 c_i0-q0 q2 c_i2+q0 q1 c_i3),
Q_j=sum_i a_ij(q2^2 c_i1-q1 q2 c_i2+(q1^2-q0 q2)c_i3),
S=sum_m Q_m^2 Q_{m+1}, T=sum_m(2Q_m Q_{m+1}+Q_{m-1}^2)P_m (indices mod 5),
R_j=q2*S*P_j+(q1*S-q2*T)*Q_j. Phi(a)=[R_0:R_1:R_2:R_3:R_4].
The R_j are homogeneous of common degree 568 before cancellation. No algebraic root is chosen.

Witness: rows b0=(1,1,0,0,1), b1=(0,1,-1,1,0), b2=(-1,-1,1,-1,-1), b3=(-1,1,-1,0,0).
At b: Delta=q2=1, (q0,q1)=(1,3), Phi(b)=[3:-2:2:-1:2].
Exact Jacobian minor d(R1/R0,R2/R0,R3/R0)/d(a00,a02,a10) = -221/27 != 0.
Hence Phi is dominant. The construction is intrinsic (isotropic planes, radical, relation, residual secant), hence G-equivariant.
Source: Tschinkel--Zhang, arXiv:2409.08392, Sec. 4. Full TeX source in this directory.